

Day 6 — Real-World AI Use Cases in Business

Course

GPT + AI Agents + RAG + n8n + API Integration + Business Automation

Lecture Duration

60 Minutes

Level

Beginner

Day 6 Main Goal

During Days 1–5, students learned:

- What Generative AI is
- How GPT and LLMs work
- Prompt engineering
- Multi-step prompting
- Structured outputs
- JSON
- Schemas
- Validation

Day 6 answers a different question:

Where should businesses actually use AI?

The purpose of today's class is to stop thinking only about technology and start thinking about:

```
Business Problem
  ↓
Process
  ↓
AI Opportunity
  ↓
Required Data
  ↓
```

Automation Design

↓

Business Result

Students should understand that successful AI projects usually begin with a **business problem**, not with:

"We want to use ChatGPT somewhere."

1. Day 6 — 60-Minute Lecture Schedule

Time	Topic
0-5 min	Day 5 review
5-10 min	What makes a good AI use case?
10-17 min	AI opportunity identification
17-25 min	Customer support and sales
25-32 min	Marketing and ecommerce
32-39 min	HR, recruitment, finance
39-45 min	Operations and management
45-51 min	Good vs bad AI automation
51-56 min	AI opportunity scoring framework
56-60 min	Mini project, quiz, homework

2. Opening Review — 0 to 5 Minutes

Ask students:

Question 1

Why do we use structured output?

Answer:

So software systems can process AI results reliably.

Question 2

What is JSON?

Answer:

A common machine-readable structured data format.

Question 3

What is a schema?

Answer:

A definition of what data should look like.

Question 4

Should GPT directly authorize a high-value financial action?

Answer:

No.

Use:

```
GPT
↓
Structured Recommendation
↓
Business Rules
↓
Authorization
↓
Human Approval if required
```

Question 5

What was the Day 5 golden rule?

AI interprets. Systems verify. Business rules decide. Tools act.

Tell students:

Today we are going to use that rule to examine entire business processes.

3. Technology First vs Business First

Show two approaches.

Bad Approach

We have ChatGPT.

↓

Where can we put it?

Better Approach

What business problem costs us time or money?

↓

Can AI help?

↓

What information does AI need?

↓

What systems must connect?

↓

What result should improve?

This is how AI consultants and automation architects should think.

4. What Is an AI Use Case?

An AI use case is a specific business problem where AI performs a useful role.

Example:

Business Problem:

500 support emails every day.

Current Process:

Employees manually read and categorize every message.

AI Use Case:
Automatically classify incoming emails.

AI Output:
category
priority
sentiment

Business Result:
Reduce manual triage work.

5. AI Use Case vs AI Feature

Explain the distinction.

Feature

"Use GPT to summarize text."

Business Use Case

"Automatically summarize every sales call and save action items into the CRM."

The second one describes:

- Problem
- Process
- Data
- AI
- Integration
- Outcome

6. Good AI Projects Solve Problems

A good project normally addresses one or more of:

Time
Cost
Errors

Slow Response
Manual Data Entry
Information Overload
Poor Customer Experience
Inconsistent Decisions
Repetitive Work
Knowledge Access

7. Common Signs of a Good AI Opportunity

Look for tasks that are:

- Repetitive
- Language-heavy
- High-volume
- Time-consuming
- Rule-supported
- Data-rich
- Currently manual
- Easy to review
- Suitable for human escalation

Examples:

Reading emails
Summarizing documents
Classifying tickets
Extracting invoice data
Drafting responses
Analyzing reviews
Searching policies
Qualifying leads

8. Tasks Less Suitable for Pure AI Automation

Be careful with tasks that involve:

- High financial impact
- Legal consequences
- Medical decisions
- Safety-critical decisions

- Irreversible actions
- Weak data
- Extremely rare edge cases
- Highly subjective judgment
- Unclear business rules

Such cases often require:

AI Assistance
+
Human Review

rather than full automation.

9. The AI Opportunity Formula

Teach:

AI Opportunity
=
High Manual Effort
+
Language / Knowledge Task
+
Repeatable Process
+
Available Data
+
Measurable Outcome

The more of these factors exist, the stronger the opportunity usually is.

10. Four Questions to Ask Before Using AI

Question 1

What is the current business problem?

Example:

Support agents spend three hours per day reading repetitive emails.

Question 2

What part requires language understanding?

Example:

Determine what the customer wants.

Question 3

What data does AI need?

Example:

- Email text
 - Customer profile
 - Order number
 - Company policy
-

Question 4

What action happens after AI?

Example:

```
Classification
↓
Routing
↓
API Lookup
↓
Response
```

11. AI Opportunity Identification — 10 to 17 Minutes

Introduce this framework:

INPUT → AI TASK → DECISION → ACTION → OUTCOME

Example:

```
INPUT:
Customer Email

AI TASK:
Classify intent

DECISION:
Billing / Shipping / Sales

ACTION:
Route ticket

OUTCOME:
Faster support
```

12. Example — Sales Lead

```
INPUT:
Lead form

AI TASK:
Extract company, budget, intent

DECISION:
High / Medium / Low quality

ACTION:
Assign sales rep
```

OUTCOME:
Sales team focuses on better leads

13. Example — Invoice

INPUT:
Invoice PDF

AI TASK:
Extract vendor, date, amount

DECISION:
Valid / Missing information

ACTION:
Send to accounting workflow

OUTCOME:
Reduce manual data entry

14. Example — Internal Knowledge

INPUT:
Employee question

AI TASK:
Understand question

DECISION:
Find relevant policy

ACTION:
Retrieve policy with RAG

OUTCOME:
Faster access to company knowledge

15. Customer Support Use Cases — 17 to 21 Minutes

Customer support is one of the easiest departments for beginners to understand.

Possible AI tasks:

- Ticket classification
- Sentiment detection
- Priority detection
- Email summarization
- Suggested replies
- FAQ answering
- Order-status assistance
- Complaint analysis
- Escalation detection
- Ticket summarization

16. Customer Support Architecture

Basic:

```
Customer Email
  ↓
  GPT
  ↓
Category
Priority
Summary
  ↓
Support Queue
```

More advanced:

```
Customer
  ↓
n8n
  ↓
AI Agent
├─ GPT
├─ RAG
```

└─ Order API
└─ CRM
└─ Ticket System

↓

Response / Human Review

17. Example — Ticket Classification

Customer:

I was charged for a subscription after cancelling.

AI output:

```
{  
  "category": "billing",  
  "priority": "high",  
  "sentiment": "negative",  
  "requires_external_data": true  
}
```

Workflow:

```
Billing  
↓  
Subscription API  
↓  
Verify cancellation  
↓  
Human review if financial adjustment is required
```

18. Customer Support Business Benefits

Possible metrics:

Average Response Time
Tickets per Agent
Resolution Time

First Contact Resolution
Escalation Rate
Customer Satisfaction

AI should be connected to measurable business outcomes.

19. Sales Use Cases — 21 to 25 Minutes

AI can help sales teams with:

- Lead qualification
 - Lead scoring
 - Lead research
 - Personalized outreach
 - Sales-call summaries
 - Follow-up drafting
 - CRM data entry
 - Objection analysis
 - Proposal drafting
 - Meeting preparation
-

20. Lead Qualification Example

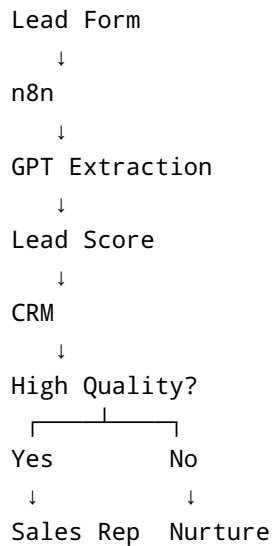
Lead form:

We are a 50-person company looking for an AI customer-service platform. Budget is around \$30,000 and we want to launch within 60 days.

AI output:

```
{  
  "company_size": 50,  
  "service_interest": "AI customer service",  
  "budget": 30000,  
  "timeline_days": 60,  
  "lead_quality": "high"  
}
```

21. Sales Automation Architecture



22. Marketing Use Cases — 25 to 28 Minutes

Possible AI tasks:

- Content ideas
- Blog outlines
- Ad copy
- Email campaigns
- Product descriptions
- Social-media drafts
- SEO research assistance
- Campaign analysis
- Review summarization
- Customer segmentation

23. Marketing Example

Input:

Product:
Vitamin C Serum

Audience:
Women aged 25–45

Benefits:
Brightening
Hydration
Lightweight

AI can generate:

Headline
Product description
Social caption
Email copy
Ad variations

But AI-generated marketing content should still follow:

- Brand rules
- Advertising policies
- Factual product claims
- Legal/compliance requirements

24. Marketing Workflow Example

```
graph TD; A[Product Database] --> B[GPT]; B --> C[Description Draft]; C --> D[Brand Rule Check]; D --> E[Human Review]; E --> F[Shopify];
```

25. Ecommerce Use Cases — 28 to 32 Minutes

AI can help with:

- Product descriptions
 - Customer support
 - Review analysis
 - Order support
 - Inventory alerts
 - Return analysis
 - Personalized recommendations
 - Product-question answering
 - Customer segmentation
 - Sales reporting
-

26. Ecommerce Support Example

Customer:

Do you have this jacket in blue XL?

GPT can understand:

```
{  
  "product": "jacket",  
  "color": "blue",  
  "size": "XL",  
  "intent": "inventory_check"  
}
```

But GPT does not know inventory.

Architecture:

```
Customer Question  
  ↓  
GPT Extraction  
  ↓  
Inventory API  
  ↓  
Live Stock
```

↓
GPT Response

27. Important Lesson

AI understands the question. The business system supplies the truth.

This pattern appears repeatedly.

28. HR Use Cases — 32 to 35 Minutes

Possible AI tasks:

- Employee FAQ
- Job-description drafting
- Resume information extraction
- Interview-question generation
- Employee survey analysis
- Onboarding assistance
- Policy search
- Training content generation
- Meeting summaries

29. HR Knowledge Assistant

Employee:

How many vacation days do I receive?

Architecture:

Question
↓
RAG
↓
HR Policy
↓
GPT

↓
Answer + Source

Important:

Do not allow the AI to invent company policies.

30. Recruitment Use Cases

AI can help:

Resume
↓
Extract:
Skills
Experience
Education
↓
Recruiter Dashboard

AI can also summarize candidate information.

However:

AI should support recruiters, not become an unreviewed final decision-maker for sensitive employment decisions.

31. Finance Use Cases — 35 to 39 Minutes

Possible AI tasks:

- Invoice extraction
 - Expense classification
 - Financial-report summarization
 - Variance explanations
 - Contract data extraction
 - Accounts-payable triage
 - Email classification
 - Financial document search
-

32. Invoice Processing Example

Invoice:

```
Vendor: Cloud Systems Inc.  
Invoice #: INV-8832  
Amount: $4,250  
Due: September 15
```

AI:

```
{  
  "vendor": "Cloud Systems Inc.",  
  "invoice_number": "INV-8832",  
  "amount": 4250,  
  "due_date": "September 15"  
}
```

Then:

```
Schema Validation  
↓  
Accounting Rules  
↓  
Approval  
↓  
Accounting System
```

33. Finance Rule

Very important:

```
AI Extraction  
≠  
Financial Approval
```

Correct architecture:

AI extracts invoice



System verifies



Business rules



Human approval



Payment

34. Operations Use Cases — 39 to 42 Minutes

Operations teams handle:

- Inventory
- Orders
- Logistics
- Suppliers
- Documents
- Processes
- Reporting
- Exceptions

AI can help with:

- Daily report summaries
- Exception detection
- Supplier email processing
- Shipment analysis
- Document extraction
- Task prioritization
- Workflow routing

35. Operations Example

Input:

Inventory Data
Orders
Supplier Emails

AI could generate:

```
{
  "risk": "inventory_shortage",
  "product": "Product A",
  "severity": "high",
  "recommended_action": "review supplier lead time"
}
```

But exact inventory quantities should come from the inventory system.

36. Management Use Cases — 42 to 45 Minutes

Executives often have an **information overload** problem.

Possible AI use cases:

- Daily executive summaries
 - Weekly KPI reports
 - Meeting summaries
 - Risk summaries
 - Customer-feedback trends
 - Sales-pipeline analysis
 - Operations reports
-

37. Executive Brief Architecture

```
CRM
+
Support
+
Sales
+
Inventory
+
Analytics
  ↓
Data Aggregation
  ↓
GPT
  ↓
```

Executive Summary

↓

Email / Slack

Example output:

Revenue:

Up 7%

Main Risk:

Product A inventory is low.

Customer Issue:

Shipping complaints increased 22%.

Recommended Attention:

Review fulfillment delays.

38. Document Processing

AI is very useful when businesses have large numbers of documents.

Examples:

- PDFs
- Contracts
- Forms
- Invoices
- Resumes
- Purchase orders
- Insurance documents
- Reports

Pattern:

Document

↓

AI Extraction

↓

Structured JSON

↓

Validation

39. Email Automation

Many organizations are effectively managed through email.

AI can:

- Classify emails
- Extract fields
- Summarize threads
- Draft replies
- Identify urgency
- Identify action items
- Route messages

Architecture:

```
graph TD
    A[Incoming Email] --> B[GPT]
    B --> C[Intent + Priority]
    C --> D[n8n]
    D --> E[Business Workflow]
```

40. Meetings

AI can transform meetings into workflow data.

Input:

Meeting Transcript

AI output:

```
{
  "summary": "...",
  "decisions": [],
  "action_items": [
    {
      "task": "Prepare proposal",
      "owner": "Sarah",
      "deadline": "Friday"
    }
  ]
}
```

Then:

```
Action Items
  ↓
Task Management System
```

41. Five Major AI Business Patterns

Students should recognize these everywhere.

Pattern 1 — Classification

```
Input
  ↓
AI
  ↓
Category
```

Examples:

- Email routing
 - Support tickets
 - Document type
-

Pattern 2 — Extraction

Unstructured Text



AI



Structured Data

Examples:

- Invoices
- Leads
- Resumes

Pattern 3 — Generation

Business Data



AI



New Content

Examples:

- Emails
- Reports
- Product descriptions

Pattern 4 — Retrieval + Generation

Question



Search Knowledge



AI Answer

This is RAG.

Pattern 5 — Decision Support

```
Business Data
  ↓
AI Analysis
  ↓
Recommendation
  ↓
Human / Business Rule
```

42. A Sixth Pattern — Agentic Action

Later in the course:

```
Goal
  ↓
AI Agent
  ↓
Choose Tool
  ↓
Perform Action
  ↓
Observe
  ↓
Next Action
```

Examples:

- Check order
- Update CRM
- Send email
- Book appointment

43. Good vs Bad AI Automation — 45 to 51 Minutes

Good Use Case

Classify 5,000 support emails every day.

Why good?

High Volume
Repetitive
Language-Based
Easy to Validate
Clear Output

44. Bad Use Case

Let GPT independently transfer \$1 million between bank accounts.

Why bad?

High Risk
Irreversible
Financial
Security-Sensitive
Requires Authorization

AI may be part of the workflow, but it should not independently control the action.

45. Good Use Case

Summarize a 50-page weekly operations report.

Why good?

- Time-consuming for humans
- Language-heavy
- Low-risk if reviewed
- Easy to compare to source

46. Weak Use Case

Use GPT to add $15 + 38$.

A normal calculator is:

Cheaper
Faster
Deterministic
More Reliable

Important principle:

Do not use AI where normal software is clearly better.

47. Good Use Case

Identify customer sentiment.

Traditional deterministic rules struggle with language variation.

AI works well.

48. Better as Traditional Code

IF amount > 500
→ Manager approval

No LLM required.

49. Hybrid Use Case

Customer requests refund.

AI:

Understand reason
Extract amount
Classify sentiment

Code:

Refund amount > \$500?

API:

Check order

Human:

Approve exception

This is often the best architecture.

50. AI vs Automation

Important distinction:

AI

Understands or generates information.

Automation

Moves information and executes workflows.

Example:

```
Email
↓
GPT
↓
Category
```

AI.

Then:

```
Category
↓
n8n
```

↓
Route Ticket

Automation.

51. AI + Automation

Together:

```
graph TD; A[Unstructured Human Input] --> B[AI]; B --> C[Structured Decision]; C --> D[Automation]; D --> E[Business Action];
```

This is the core of the course.

52. Human-in-the-Loop

A mature AI workflow should ask:

Which decisions can be automated?

and:

Which decisions require humans?

Example:

```
graph TD; A[AI Draft] --> B[Risk Level]; B --> C[Low]; B --> D[High]; C --> E[ ]; D --> F[ ];
```

Auto	Human
Send	Review

53. Automation Levels

Teach three levels.

Level 1 — AI Assistant

```
Human
↓
AI Suggestion
↓
Human Action
```

Example:

GPT drafts an email.

Level 2 — AI-Assisted Automation

```
Input
↓
AI
↓
Workflow
↓
Human Approval
↓
Action
```

Level 3 — Automated AI Workflow

```
Input
↓
AI
↓
```

Validation
↓
Business Rules
↓
Automatic Action

Use Level 3 only where risk is controlled.

54. AI Opportunity Scoring — 51 to 56 Minutes

Students now need a way to evaluate potential projects.

Use six dimensions.

AI Opportunity Score

Business Value
Automation Potential
AI Suitability
Data Availability
Implementation Difficulty
Risk

Score each from:

1 to 5

55. Business Value

How valuable is solving the problem?

1

Very little impact.

5

Major impact on:

- Revenue
 - Cost
 - Speed
 - Customer experience
-

56. Automation Potential

How repetitive is the process?

1

Rare and unique.

5

Highly repetitive and standardized.

57. AI Suitability

Does the task involve:

- Language?
- Documents?
- Classification?
- Extraction?
- Generation?
- Semantic understanding?

5

Excellent fit for AI.

58. Data Availability

Does the organization have the needed information?

Examples:

Emails
Documents
CRM records
Policies
Order data

Without good data, the AI project becomes harder.

59. Implementation Difficulty

1

Very easy.

Example:

Summarize email

5

Very difficult.

Example:

Multi-agent system connected to 12 enterprise systems

When ranking opportunities, lower difficulty is usually preferable for early projects.

60. Risk

Low risk

- Draft content
- Internal summaries
- Classification

High risk

- Financial actions
- Legal decisions

- Safety decisions
- Security permissions

Higher risk requires stronger controls.

61. AI Opportunity Table

Example:

Opportunity	Value	Automation	AI Fit	Data	Difficulty	Risk
Email classification	4	5	5	5	2	1
Auto-refund \$10k	4	4	2	4	3	5
Review summarization	4	5	5	5	2	1
Sales lead scoring	5	4	4	4	3	2

Students can quickly see which projects are attractive.

62. Best Beginner Projects

Prefer:

```
High Value
+
High Repetition
+
Good AI Fit
+
Available Data
+
Low/Medium Difficulty
+
Low Risk
```

Example:

Customer email classification.

63. Bad First AI Project

Avoid:

High Risk
+
Poor Data
+
Complex Integrations
+
No Defined Outcome

Example:

Fully autonomous corporate financial controller.

64. AI ROI Basics

Students should understand return on investment.

Suppose:

Employees spend:
100 hours/month

Cost:
\$30/hour

Monthly cost:

$100 \times \$30$
=
\$3,000

Suppose automation reduces manual work by 60%.

Saved:

60 hours/month

Approximate labor capacity saved:

\$1,800/month

65. AI ROI Is Not Only Labor Savings

Benefits may include:

- Faster response
- Better consistency
- Increased conversion
- Reduced errors
- Better customer satisfaction
- Faster onboarding
- Better access to knowledge

66. Important ROI Warning

Do not claim:

AI will reduce cost by 90%.

without evidence.

Instead:

Current baseline
↓
Pilot
↓
Measure
↓
Compare

Professional AI projects require measurement.

67. Baseline

Before automation:

Average Response Time:
8 hours

Tickets per Agent:
40/day

After pilot:

Average Response Time:
3 hours

Tickets per Agent:
60/day

Now there is measurable evidence.

68. Business Process Mapping

Before building AI, document the current process.

Example:

```
Customer Email
  ↓
Agent Opens Email
  ↓
Reads Message
  ↓
Finds Order
  ↓
Checks Policy
  ↓
Writes Reply
  ↓
Updates CRM
```

69. Identify AI Opportunities Inside the Process

```
graph TD; A[Customer Email] --> B[AI Classification]; B --> C[AI Extraction]; C --> D[Order API]; D --> E[RAG Policy Search]; E --> F[AI Response Draft]; F --> G[Human Approval]; G --> H[CRM Update];
```

Not every step must use AI.

70. Important Principle

Automate the process, not just the prompt.

A prompt alone may save 30 seconds.

A complete workflow may save 10 minutes.

71. Business Automation Architecture

Use four layers:

```
graph TD; A[INPUT] --> B[INTELLIGENCE]; B --> C[INTEGRATION]; C --> D[ACTION];
```

Example:

```
INPUT
Customer Email

INTELLIGENCE
GPT Classification

INTEGRATION
Order API

ACTION
Send Response
```

72. Add Knowledge

```
INPUT
↓
GPT
↓
RAG
↓
API
↓
Workflow
↓
ACTION
```

This becomes a full AI business system.

73. AI Solution Design Template

Students should use this for every use case.

1. Business Problem

What is wrong today?

2. Current Workflow

How do employees handle it?

3. Input

What data enters?

4. AI Task

What should AI do?

5. External Data

What additional facts are required?

6. Business Rules

What deterministic rules exist?

7. Action

What happens next?

8. Human Review

When is approval required?

9. Outcome

What should improve?

10. Metric

How will success be measured?

74. Example — Customer Support

Business Problem

Support team spends too much time reading tickets.

Input

Customer email.

AI Task

Classify:

```
billing
shipping
returns
sales
```

External Data

None for initial classification.

Business Rule

High-priority billing issues go to senior agents.

Action

Route ticket.

Metric

Time spent on triage.

75. Example — Order Status Agent

Business Problem

Customers repeatedly ask:

Where is my order?

AI Task

Extract order ID.

API

Order management system.

RAG

Shipping policy.

Action

Generate response.

Human Review

Required for unusual disputes.

Metrics

- Tickets resolved automatically
 - Response time
 - Escalation rate
-

76. Example — Sales Lead Automation

Business Problem

Sales reps manually review every incoming lead.

Input

Lead form.

AI Task

Extract and classify.

Output

```
{  
  "budget": 20000,  
  "timeline": "30 days",  
}
```

```
"intent": "high",  
"lead_score": "high"  
}
```

Action

Assign high-quality lead.

Metrics

- Lead response time
 - Qualified lead rate
 - Conversion rate
-

77. Example — Internal HR Assistant

Problem

HR answers the same policy questions repeatedly.

AI

Understand employee question.

RAG

Search employee handbook.

Output

Grounded answer.

Metrics

- HR tickets reduced
 - Employee response time
-

78. Example — Invoice Automation

Problem

Accounting staff manually enter invoice information.

AI

Extract invoice fields.

Validation

Check required fields.

Business Rule

Amount > \$5,000 requires manager approval.

Action

Send to accounting system.

Metric

Invoice processing time.

79. Mini Project — AI Opportunity Audit

Students choose one business.

Suggested examples:

- Ecommerce brand
 - Restaurant
 - Hotel
 - Recruitment agency
 - Marketing agency
 - Software company
 - Logistics company
 - Real-estate agency
-

80. Step 1 — Select Business

Example:

Business:
Online Fashion Store

81. Step 2 — List Business Departments

Example:

Customer Support
Sales
Marketing
Operations
Inventory
Finance
Management

82. Step 3 — Find 10 Manual Tasks

Example:

1. Read customer emails
 2. Answer order questions
 3. Write product descriptions
 4. Analyze reviews
 5. Qualify wholesale leads
 6. Process invoices
 7. Summarize sales reports
 8. Monitor inventory issues
 9. Answer staff policy questions
 10. Create marketing emails
-

83. Step 4 — Convert Them to AI Opportunities

Example:

Manual Task	AI Opportunity
Read emails	Ticket classification
Answer order questions	AI support agent
Analyze reviews	Sentiment analysis
Product descriptions	Content generation
Wholesale leads	Lead qualification

84. Step 5 — Identify AI Type

Use categories:

Classification
Extraction
Generation
RAG
Agent
Analysis

Example:

Opportunity	AI Type
Support routing	Classification
Invoice processing	Extraction
Product description	Generation
Employee policy assistant	RAG
Order assistant	Agent

85. Step 6 — Required Technology

Example:

Use Case	GPT	RAG	API	n8n	Agent
Ticket classification	✓			✓	

Use Case	GPT	RAG	API	n8n	Agent
Order assistant	✓	✓	✓	✓	✓
Product description	✓			✓	
Policy assistant	✓	✓			
Lead automation	✓		✓	✓	

Students begin seeing how course technologies connect.

86. Step 7 — Score Opportunities

Use:

```
Value
AI Fit
Automation Potential
Data
Difficulty
Risk
```

87. Example Score

Customer Ticket Classification

```
Value: 4/5
AI Fit: 5/5
Automation: 5/5
Data: 5/5
Difficulty: 2/5
Risk: 1/5
```

This is an excellent beginner project.

88. Example Score

Autonomous Refund Agent

```
Value: 4/5
AI Fit: 3/5
Automation: 4/5
Data: 4/5
Difficulty: 4/5
Risk: 5/5
```

Not ideal as the first autonomous project.

89. Step 8 — Choose Top 3

Pick opportunities with:

```
High Value
High AI Fit
Low Risk
Reasonable Difficulty
```

Example:

1. Ticket classification
 2. Review analysis
 3. Lead qualification
-

90. Step 9 — Design Architecture

Example:

```
New Lead
  ↓
n8n
  ↓
GPT
  ↓
Structured JSON
```

↓
Lead Score
↓
CRM
↓
Sales Notification

91. Step 10 — Define Success Metrics

Example:

Current lead review time:
15 minutes

Target:
3 minutes

Current first response:
12 hours

Target:
30 minutes

92. Mini Project Deliverable

Students should produce:

AI Opportunity Audit

with:

Business Description

10 Manual Tasks

10 AI Opportunities

AI Type

Required Technology

Value Score

Difficulty Score

Risk Score

Top 3 Opportunities

Architecture for Best Opportunity

Success Metrics

93. Day 6 Exercise — Good or Bad AI Use Case?

Ask students to classify each.

Example 1

Summarize 500 customer reviews.

Good

Why:

Language-heavy
High volume
Low risk

Example 2

Let GPT calculate payroll taxes without verification.

Weak / Risky

Better:

Payroll software calculates
AI explains or summarizes

Example 3

Classify support emails.

Good

Example 4

Let AI approve million-dollar contracts automatically.

Bad autonomous use case

Human/legal review is required.

Example 5

Extract invoice fields.

Good

Example 6

Use AI instead of a calculator for simple arithmetic.

Poor use of AI

94. Day 6 Quiz

Question 1

What should come first?

- A. Choose AI tool
- B. Identify business problem

Answer: B

Question 2

Which is a good AI use case?

- A. Customer-email classification
- B. Deterministically adding two numbers

Answer: A

Question 3

Which task is best for traditional code?

- A. Sentiment analysis
- B. Refund amount > \$500 rule

Answer: B

Question 4

What is a hybrid workflow?

- A. AI does everything
- B. AI + code + tools + human/workflow controls

Answer: B

Question 5

What is an important AI opportunity characteristic?

- A. Repetitive manual process
- B. Completely undefined business process

Answer: A

Question 6

Should AI automatically execute every recommended action?

Answer: No

Question 7

Which technology provides company knowledge?

- A. RAG
- B. Calculator

Answer: A

Question 8

Which technology connects software systems?

- A. API
- B. Prompt

Answer: A

Question 9

Which tool will later orchestrate workflows in this course?

Answer: n8n

Question 10

How should we measure an AI project's success?

Answer:

With business metrics before and after implementation.

95. True or False

Every repetitive task requires AI.

False

Some repetitive tasks are better handled with normal automation.

True

AI works well with unstructured language.

True

AI should replace all deterministic business rules.

False

APIs provide live system information.

True

RAG can provide company knowledge.

True

Human approval can be part of an AI automation.

True

AI ROI should be measured.

True

The most technically complex AI project is always the best one.

False

96. Day 6 Vocabulary

AI Use Case

A business problem where AI provides useful functionality.

Business Process

A sequence of steps used to complete business work.

AI Opportunity

A process or task that may benefit from AI.

Automation Potential

How much of the process can be automated.

AI Suitability

How appropriate AI is for the task.

ROI

Return on Investment.

Baseline

The current performance measurement before implementing AI.

Human-in-the-Loop

A system in which humans review or approve some AI decisions.

Hybrid System

A system combining AI, deterministic software, tools, and human controls.

KPI

Key Performance Indicator.

97. Day 6 Core Framework

Students should memorize:

```
Problem
↓
Process
↓
Input
↓
AI Task
↓
Data / Knowledge
↓
Business Rule
↓
Action
↓
Human Review
↓
Outcome
↓
Metric
```

98. The AI Solution Stack

Students should start thinking in layers:

```
BUSINESS PROBLEM
↓
AI INTELLIGENCE
↓
GPT
Agents
RAG
↓
INTEGRATION
↓
API
Database
↓
ORCHESTRATION
↓
n8n
↓
BUSINESS ACTION
```

99. Day 6 Golden Rule

Write prominently:

Do not automate technology. Automate business problems.

Another useful version:

Start with the workflow, not the AI model.

100. Ten Questions AI Consultants Should Ask Clients

Students should begin learning client discovery.

- 1. What process currently takes the most employee time?**
- 2. Which tasks are repeated every day?**
- 3. Which tasks involve reading or writing large amounts of text?**
- 4. What mistakes happen frequently?**
- 5. What information do employees search for repeatedly?**
- 6. Which systems contain the required data?**
- 7. What decisions have strict business rules?**
- 8. Which actions require human approval?**
- 9. How many times per day/week/month does this process occur?**
- 10. How will we know if automation succeeded?**

These are often more valuable than asking:

Which AI model do you want?

101. Outsourcing Job Perspective

When a client says:

"We need an AI chatbot."

Do not immediately build a chatbot.

Ask:

What should the chatbot solve?

What information does it need?

Which systems does it connect to?

What actions may it perform?

Which actions require human approval?

What outcome should improve?

102. Weak Freelancer Response

I can build a ChatGPT bot using OpenAI.

103. Strong AI Architect Response

I would first map your current support workflow, identify which requests can be resolved automatically, connect the AI to your knowledge base and order system, define escalation rules, and measure the impact on response and resolution time.

That difference is how students move toward AI outsourcing expertise.

104. Homework Assignment 1 — AI Opportunity Audit

Choose one business.

Identify:

10 repetitive tasks
10 possible AI use cases
AI type
Required integrations
Expected business outcome

105. Homework Assignment 2 — Score Each Opportunity

Use:

Factor	Score
Business Value	1-5
Automation Potential	1-5
AI Suitability	1-5
Data Availability	1-5
Difficulty	1-5
Risk	1-5

Choose your top three opportunities.

106. Homework Assignment 3 — Design One Architecture

Choose your best opportunity.

Use:

```
Input
↓
GPT
↓
RAG if needed
↓
API if needed
↓
Business Rule
↓
n8n
↓
Human / Action
```

107. Homework Assignment 4 — Define Metrics

For your project, create:

```
Current Metric
Target Metric
```

Example:

```
Current:
Support response = 6 hours

Target:
Support response < 1 hour
```

Create at least three metrics.

108. Homework Assignment 5 — AI vs Automation

For each of your 10 tasks, decide:

AI
Traditional Automation
Hybrid
Human

Example:

Task	Best Approach
Analyze customer sentiment	AI
Check amount > \$500	Code
Answer order-status request	Hybrid
Approve sensitive refund	Human/Hybrid

109. Interview-Style Questions

What makes a good AI use case?

A task with meaningful business value, high repetition, language or knowledge complexity, accessible data, measurable results, and manageable risk.

Why should AI projects start from the business process?

Because the technology should solve a measurable business problem rather than exist for its own sake.

What is a hybrid AI workflow?

A workflow combining LLMs with deterministic code, APIs, business rules, and possibly human review.

What is human-in-the-loop?

A design where humans approve or review selected AI decisions.

What does ROI mean?

Return on Investment.

How can an AI project be evaluated?

Compare baseline metrics with post-implementation metrics.

110. Day 6 Learning Outcome Checklist

Students should be able to:

- ☐ Explain what an AI use case is
- ☐ Distinguish an AI feature from a business use case
- ☐ Identify repetitive business tasks
- ☐ Recognize language-heavy AI opportunities
- ☐ Identify tasks better suited to code
- ☐ Explain hybrid automation
- ☐ Explain human-in-the-loop
- ☐ Identify AI opportunities in customer support
- ☐ Identify AI opportunities in sales
- ☐ Identify AI opportunities in marketing
- ☐ Identify AI opportunities in ecommerce
- ☐ Identify AI opportunities in HR
- ☐ Identify AI opportunities in finance
- ☐ Identify AI opportunities in operations
- ☐ Score AI opportunities
- ☐ Estimate simple ROI
- ☐ Define success metrics
- ☐ Design a high-level business AI architecture

111. Day 6 Final Mental Model

Instead of:

```
GPT
↓
What can we do with it?
```

students should now think:

```
Business Problem
↓
Current Workflow
↓
Where is the manual pain?
↓
Can AI help?
↓
What data is required?
```

↓
What AI task?
↓
What API / RAG / n8n?
↓
What human control?
↓
What measurable business result?

112. Day 6 Final Takeaway

The most important lesson today is:

The best AI expert is not the person who knows the most AI tools. It is the person who can identify where AI creates measurable business value.

Students are beginning to move from:

AI User

to:

AI Solution Designer

The progression now looks like:

Understand AI
↓
Understand GPT
↓
Prompt Correctly
↓
Structure Data
↓
Understand Business Problems
↓
Design AI Solutions
↓
Build Automation

Preview — Day 7

Day 7 — Week 1 Mini Project: AI Customer Support Assistant

Day 7 will combine everything from Days 1–6.

Students will build the conceptual design for a complete:

AI Customer Support Triage Assistant

Input:

Customer Message

AI will return:

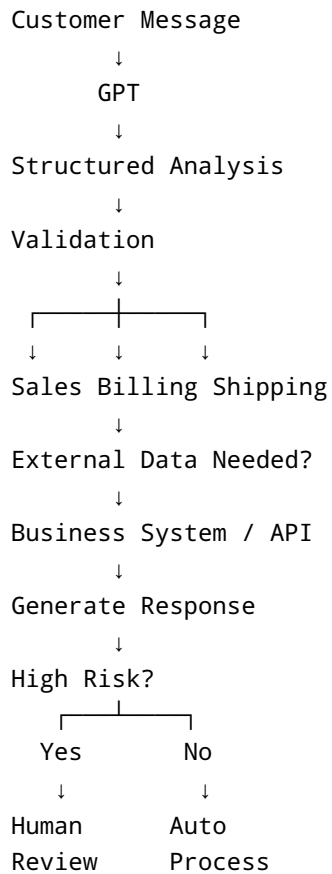
```
{
  "summary": "",
  "category": "",
  "sentiment": "",
  "priority": "",
  "customer": {},
  "entities": {},
  "missing_information": [],
  "requires_external_data": false,
  "requires_human_review": false,
  "recommended_action": "",
  "draft_response": ""
}
```

The Day 7 project will include:

- Business problem definition
- Prompt design
- Prompt decomposition
- Structured JSON output
- Schema design
- Validation rules
- Test cases
- Error handling

- Prompt-injection test
- Business rules
- Human-review logic
- Architecture diagram
- Basic ROI calculation
- Portfolio case-study format

Architecture:



Day 7 will therefore serve as the **first complete mini-project of the 30-day course**.